Firewall Implementation using IPFire



Presentation

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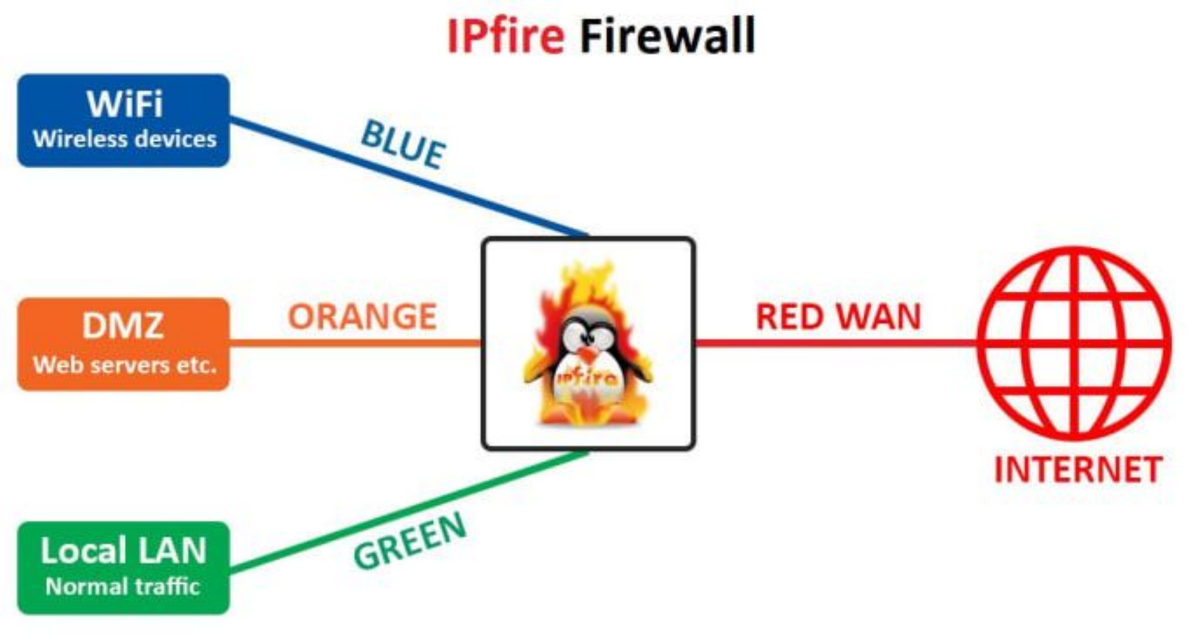
3 Results

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Intro + Applying MoSCow Method



Introduction



- Secure, Fast and Versatile
- suitable for Data Centers, Businesses and Home Networks
- offers advanced features
  - NAT and DNS Security

- Will be using **Four Interfaces:**
  - WAN
  - LAN
  - DMZ
  - WLAN

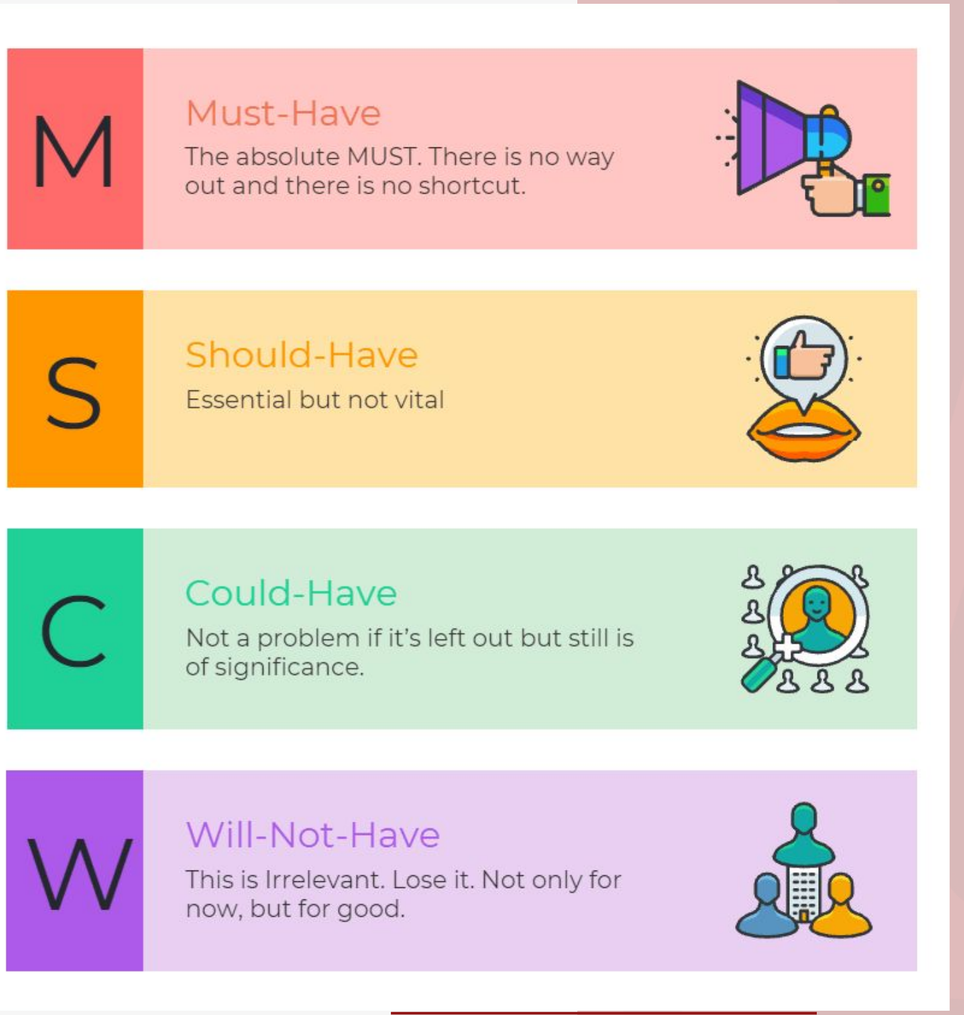
MoSCow Method

Essential Features

- Firewall Rules & Packet Filtering
- IDS / IPS
- SSH and Web Interfaces

Important Features

- VPN Configuration
- Web Proxy with Content Filtering
- Quality of Service (QoS)



MoSCow Method

M — Must-Have

The absolute MUST. There is no way out and there is no shortcut.



S — Should-Have

Essential but not vital



C — Could-Have

Not a problem if it's left out but still is of significance.



W — Will-Not-Have

This is Irrelevant. Lose it. Not only for now, but for good.



Desired Features

- Logging & Monitoring

  Email Alerts
- NAT Configuration

Least Critical Features

- Load Balancing &

  Failover
- IPv6 Configuration



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Procedures to Implement

Essential Features

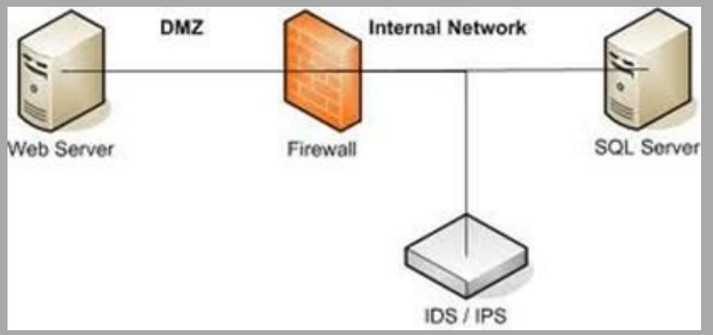
ISO File

- Configure Network Setup
- Assign NIC for each network with address



Firewall Rules

- Select protocol and specify ports
- Set rule action for internal network



IDS / IPS

- Determine which network needs traffic analysis
- Add and customize rule set

SSH & Web Interface

- Password-based Authentication
- Enable TCP forwarding
- Set port 22 for default SSH



Important Features

VPN Configuration

- Set IP settings
- Allow UDP port 1194



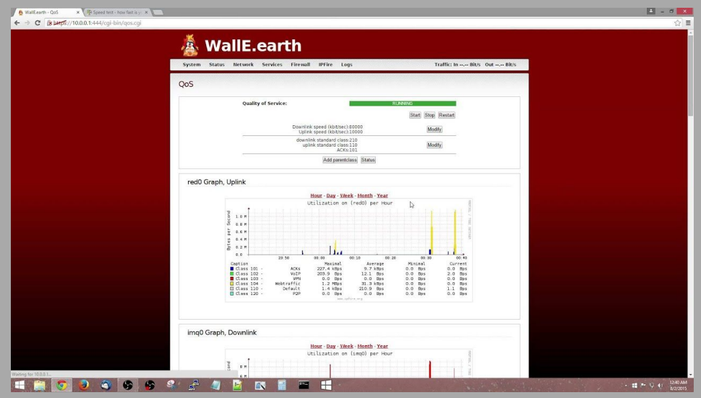
Web Proxy

- Enable URL Filtering
- Configure domains for to block restrictions
- Enable logging to track



Quality of Service

- Assign Priority
- Specify Destination Port and Source Port



Results and Expectations

- Improved security through controlled traffic and intrusion detection
- Content control that aligns with business policies.
- Secure remote access using SSH and VPN
- Public access to DMZ web server
- Ping:

  Red (WAN) >> Orange (DMZ)

  Blue (WLAN) >> Orange (DMZ)

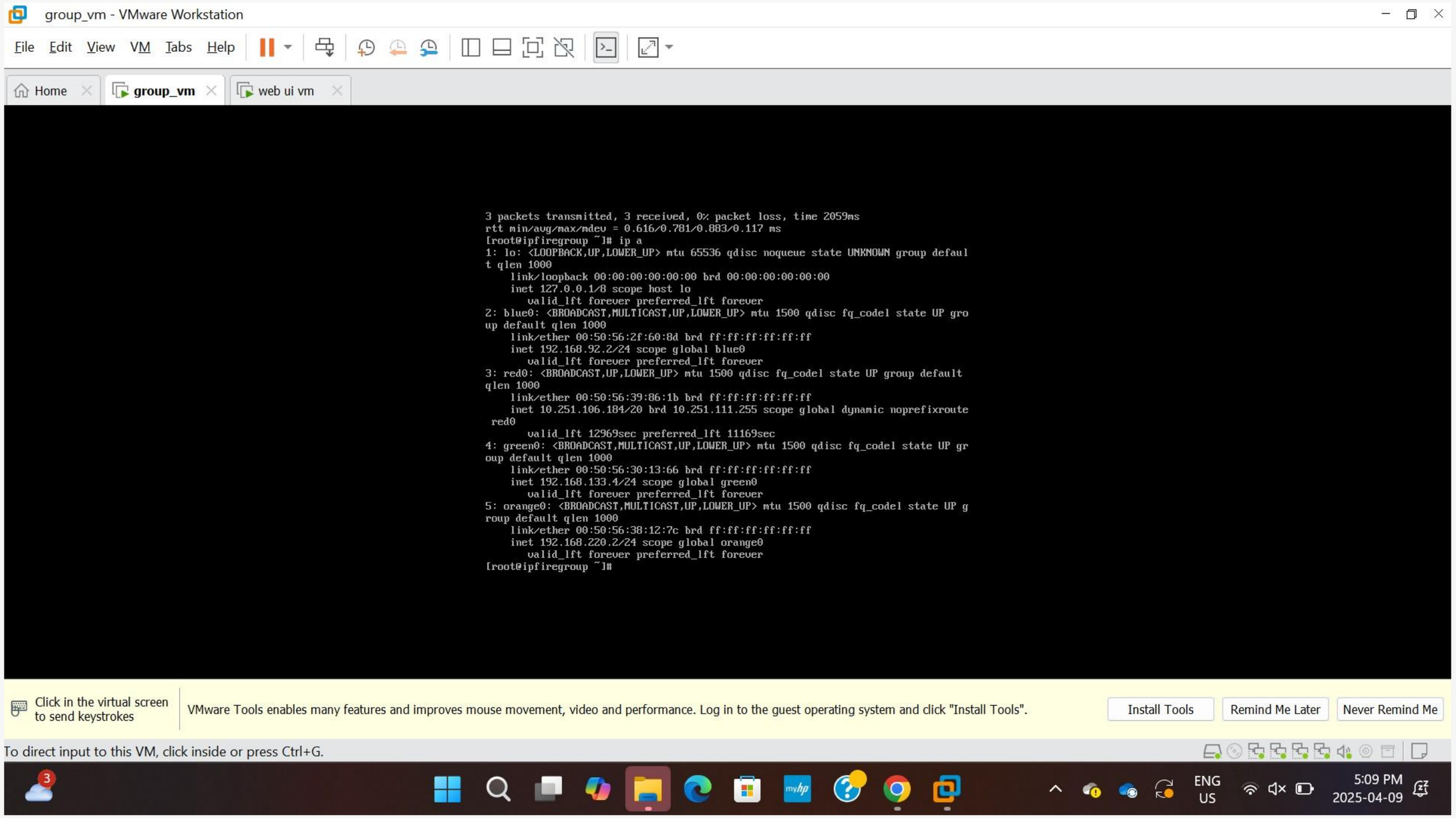
  Blue (WLAN) >> Green (LAN)

Default zone ruleset

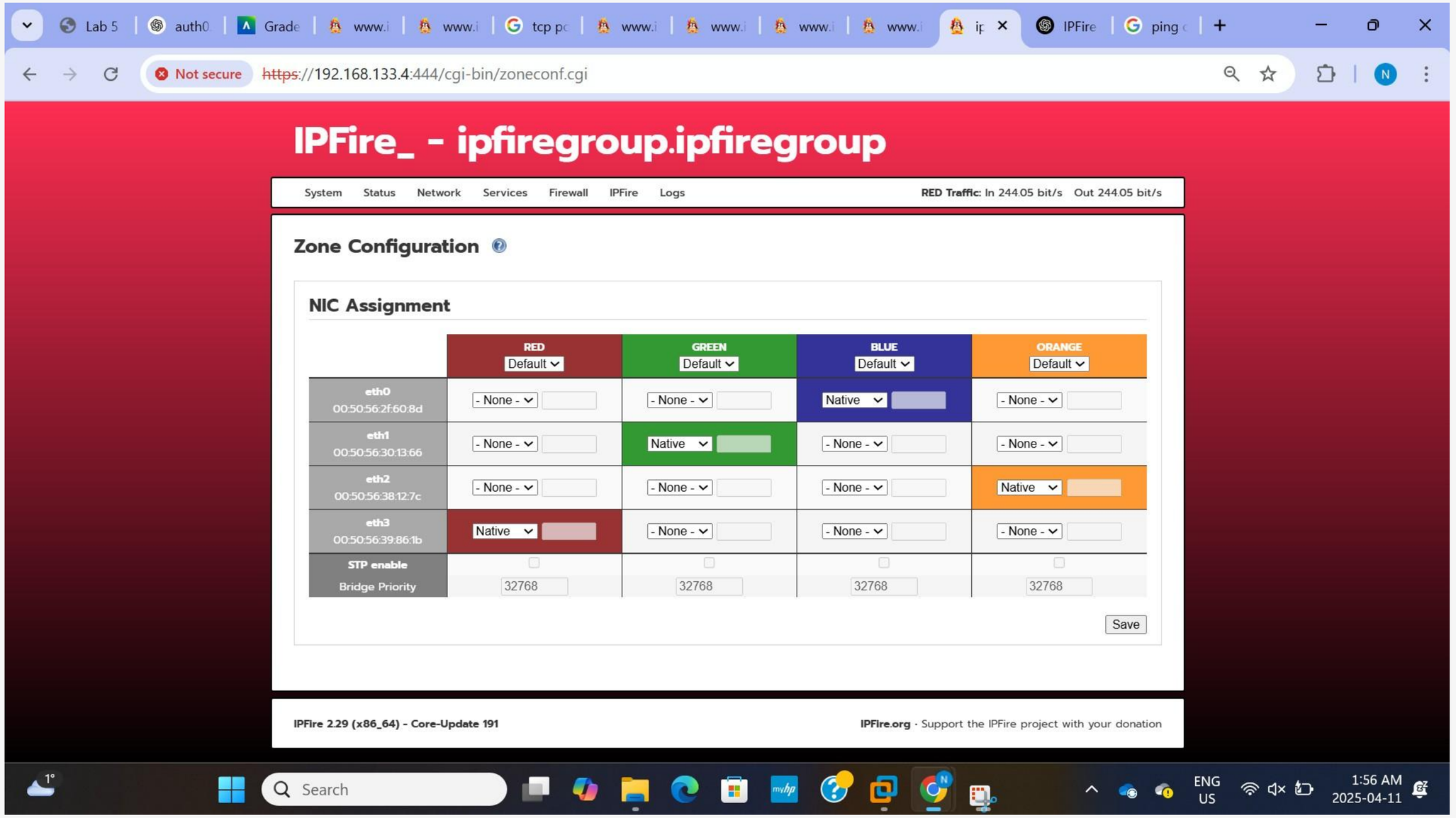
IPFire comes with a default ruleset which restricts the traffic between the individual network zones. The following table shows this limitations:

****	Direction	****	Status
Red	->	**Firewall**	Closed, Use external access
Red	->	**Orange**	Closed. Use port forwarding
Red	->	**Blue**	Closed. Use port forwarding or VPN
Red	->	**Green**	Closed. Use port forwarding or VPN
Orange	->	**Firewall**	Closed, no DNS nor DHCP[1] for **Orange**
Orange	->	**Red**	Open
Orange	->	**Blue**	Closed, use DMZ pinholes
Orange	->	**Green**	Closed, use DMZ pinholes
Blue	->	**Firewall**	Closed, use Blue Access
Blue	->	**Red**	Closed, use Blue Access
Blue	->	**Orange**	Closed, use Blue Access
Blue	->	**Green**	Closed[2], use Creating a Blue to Green Pinhole or VPN
Green	->	**Firewall**	Open
Green	->	**Red**	Open

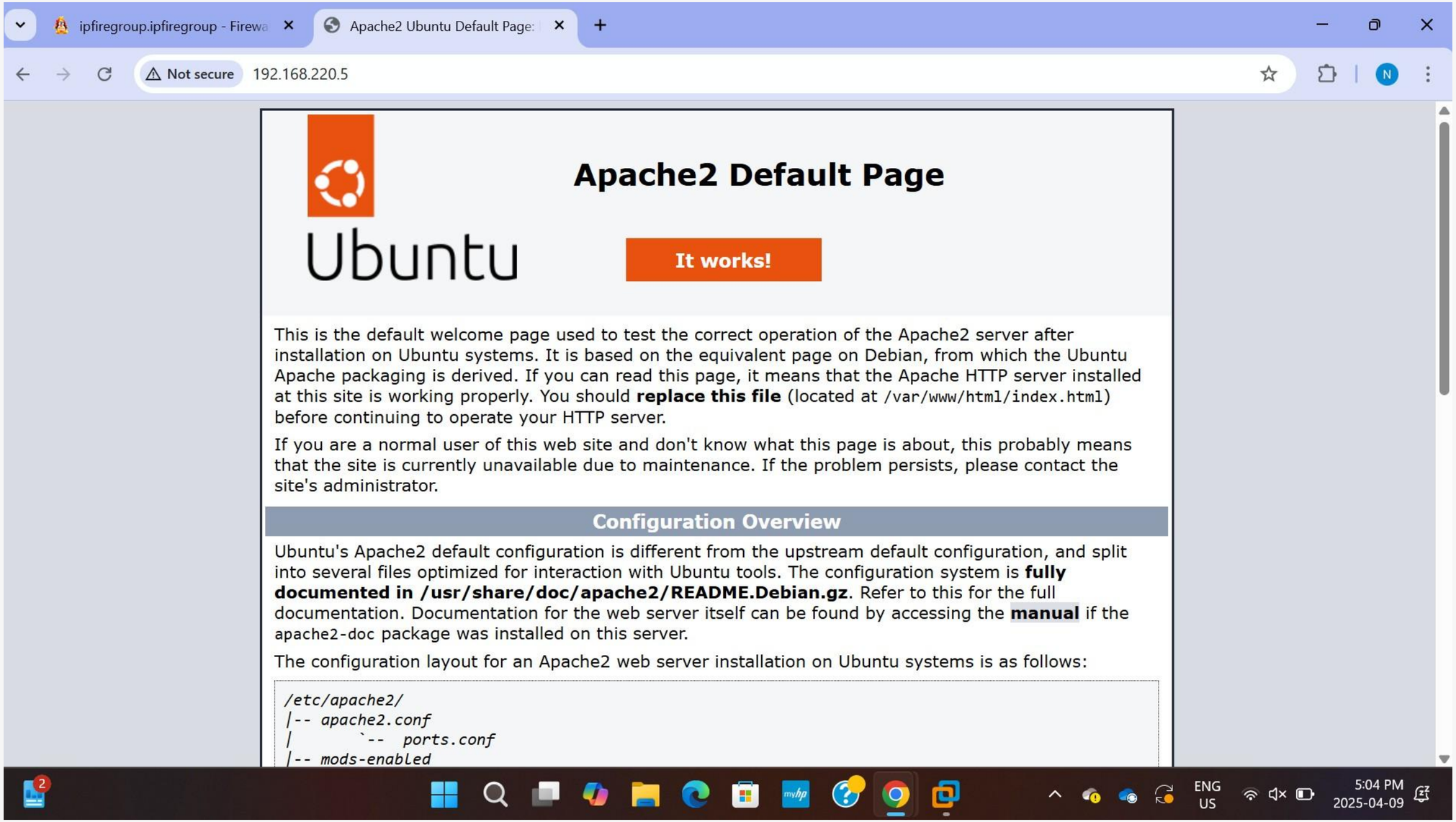
ip addresses on vm for all interfaces



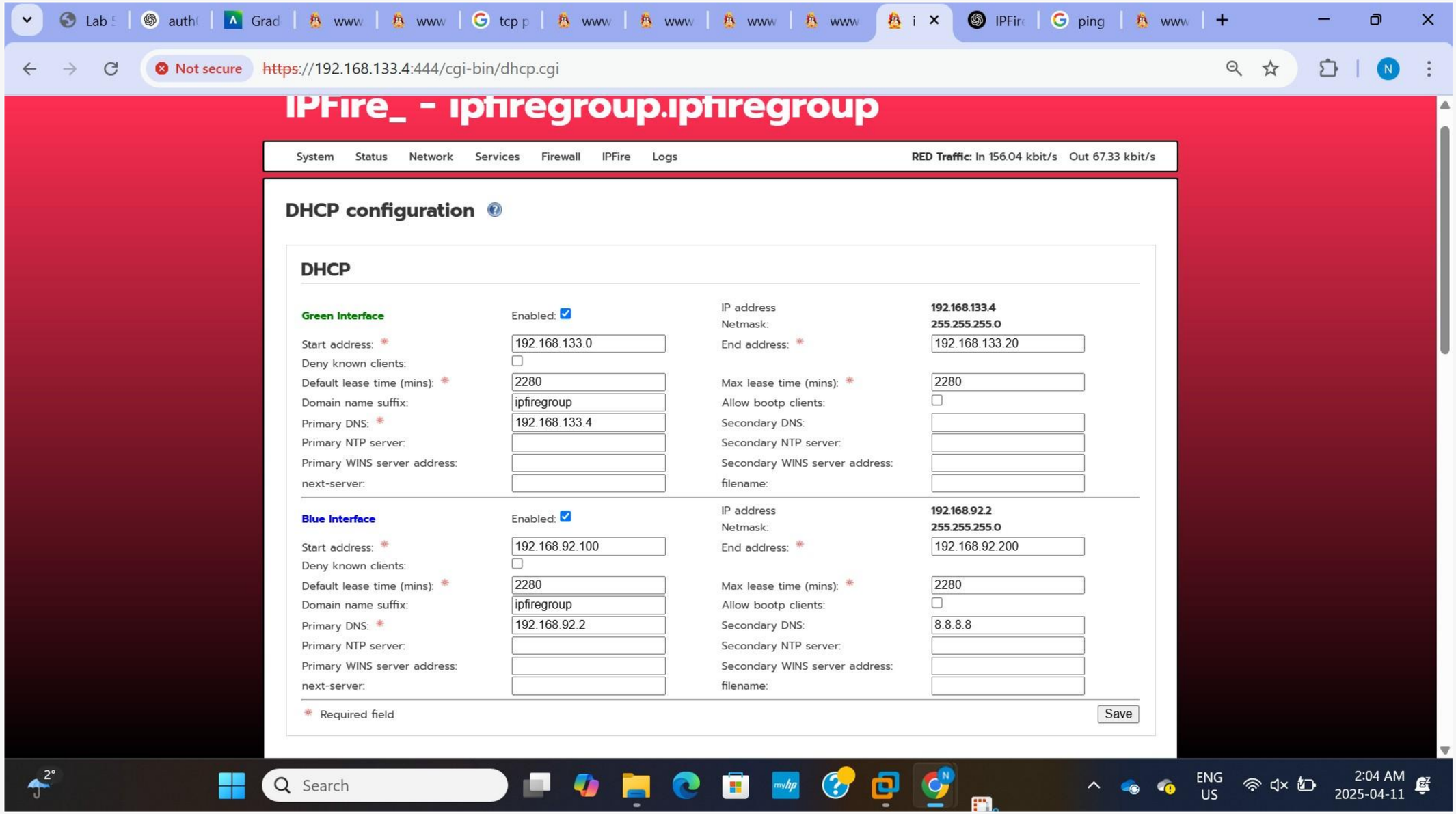
zone configuration for each interface



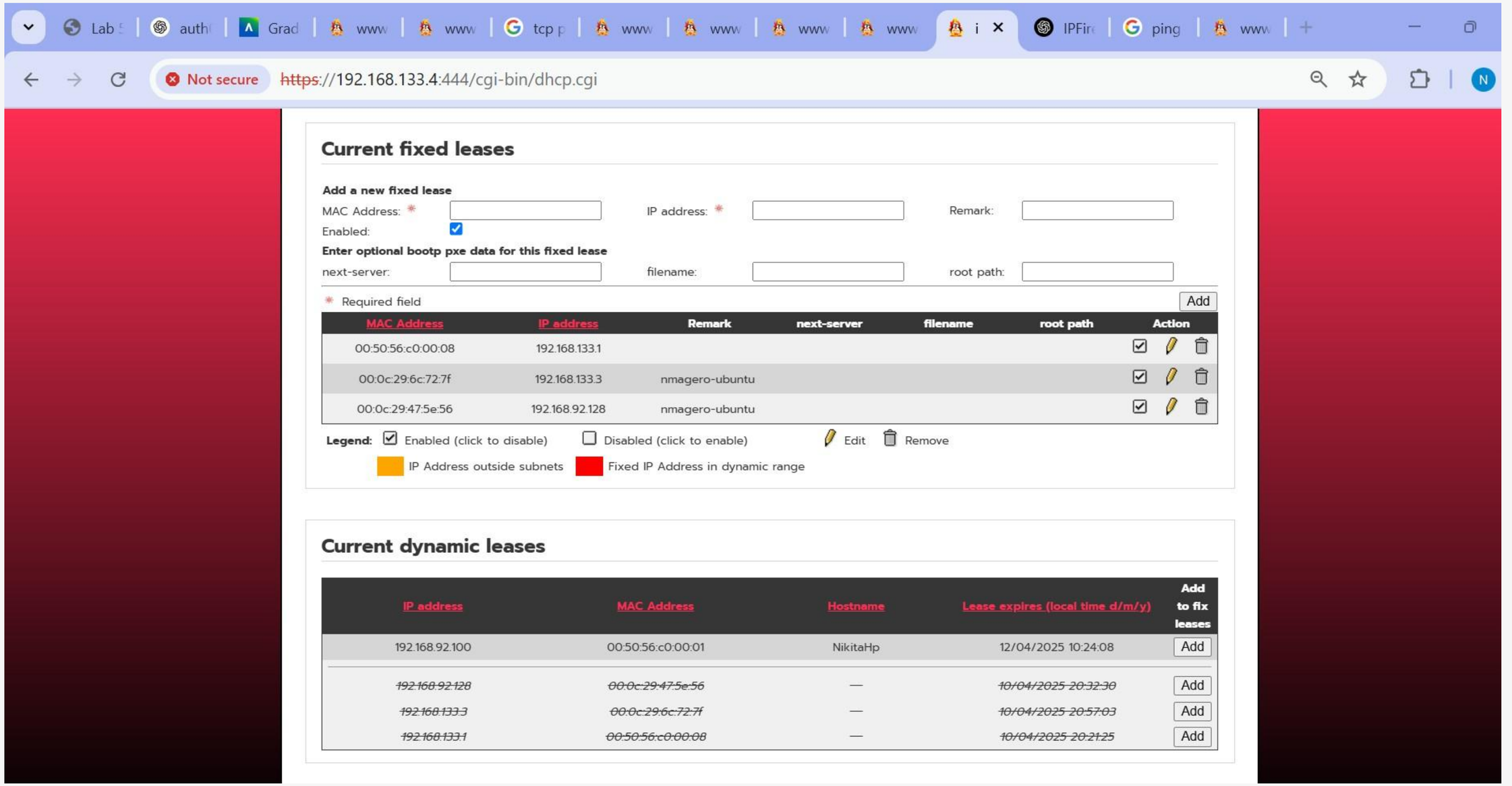
Orange Configuration



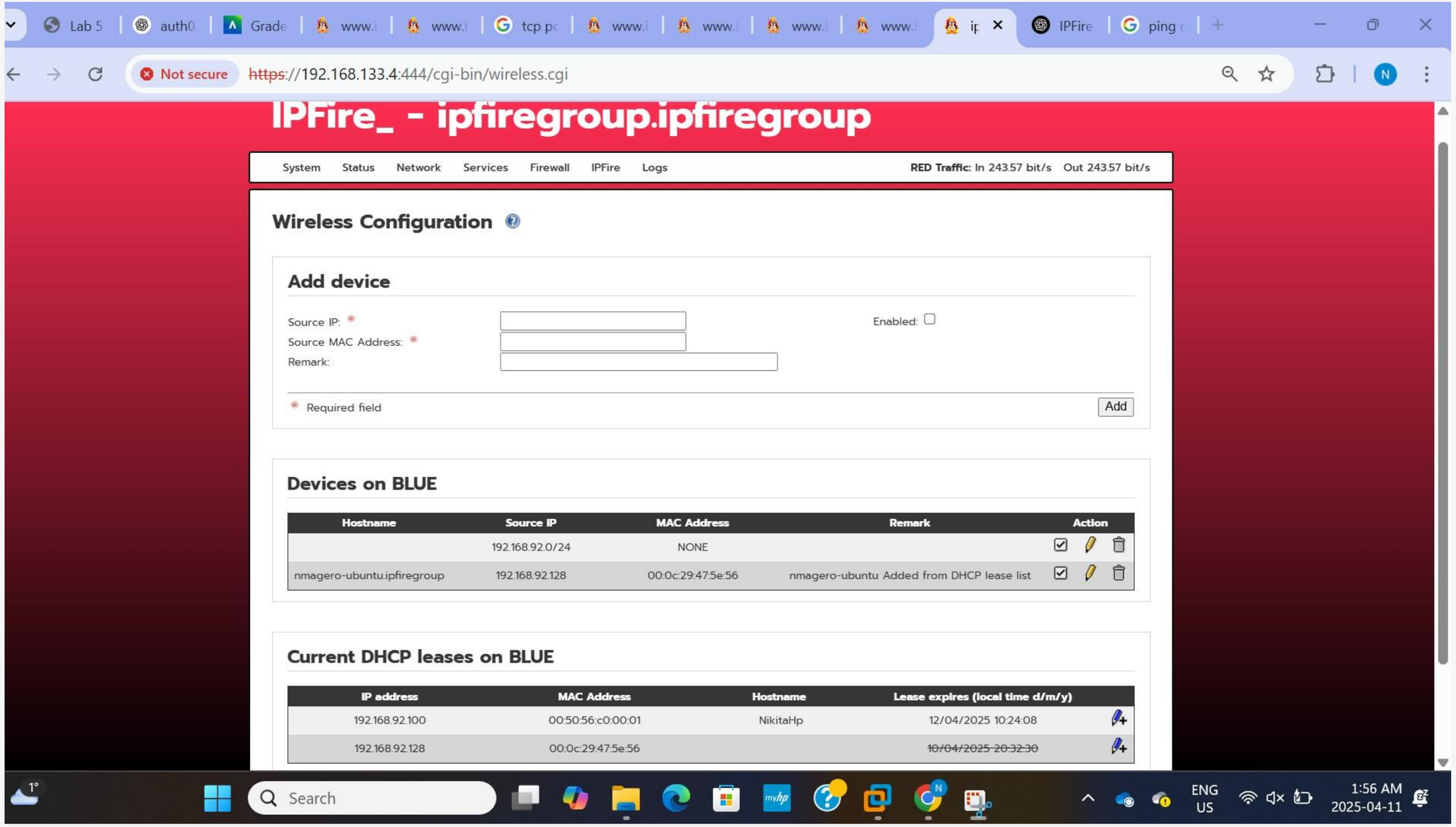
blue configuration



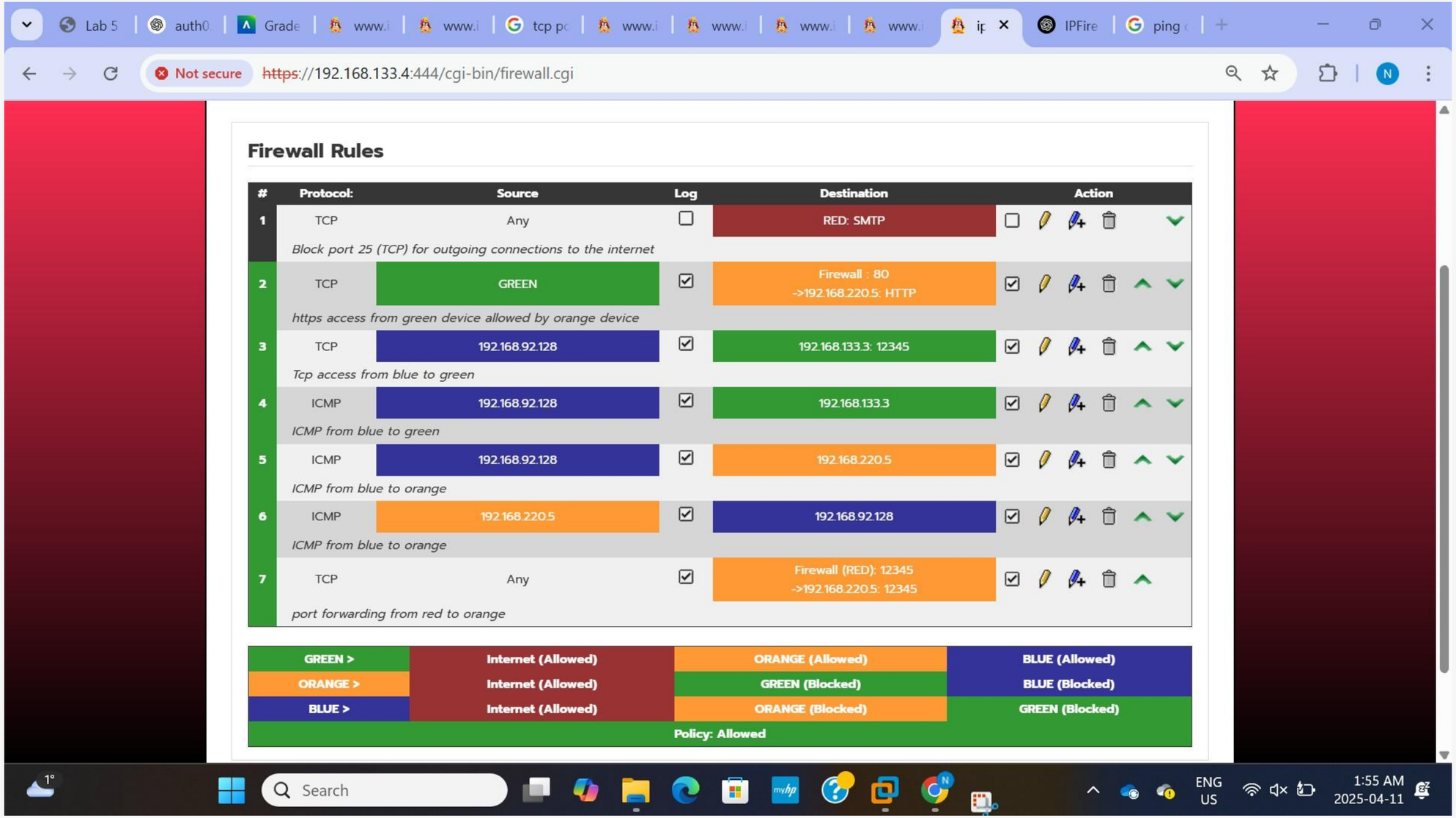
dhcp fixed leases



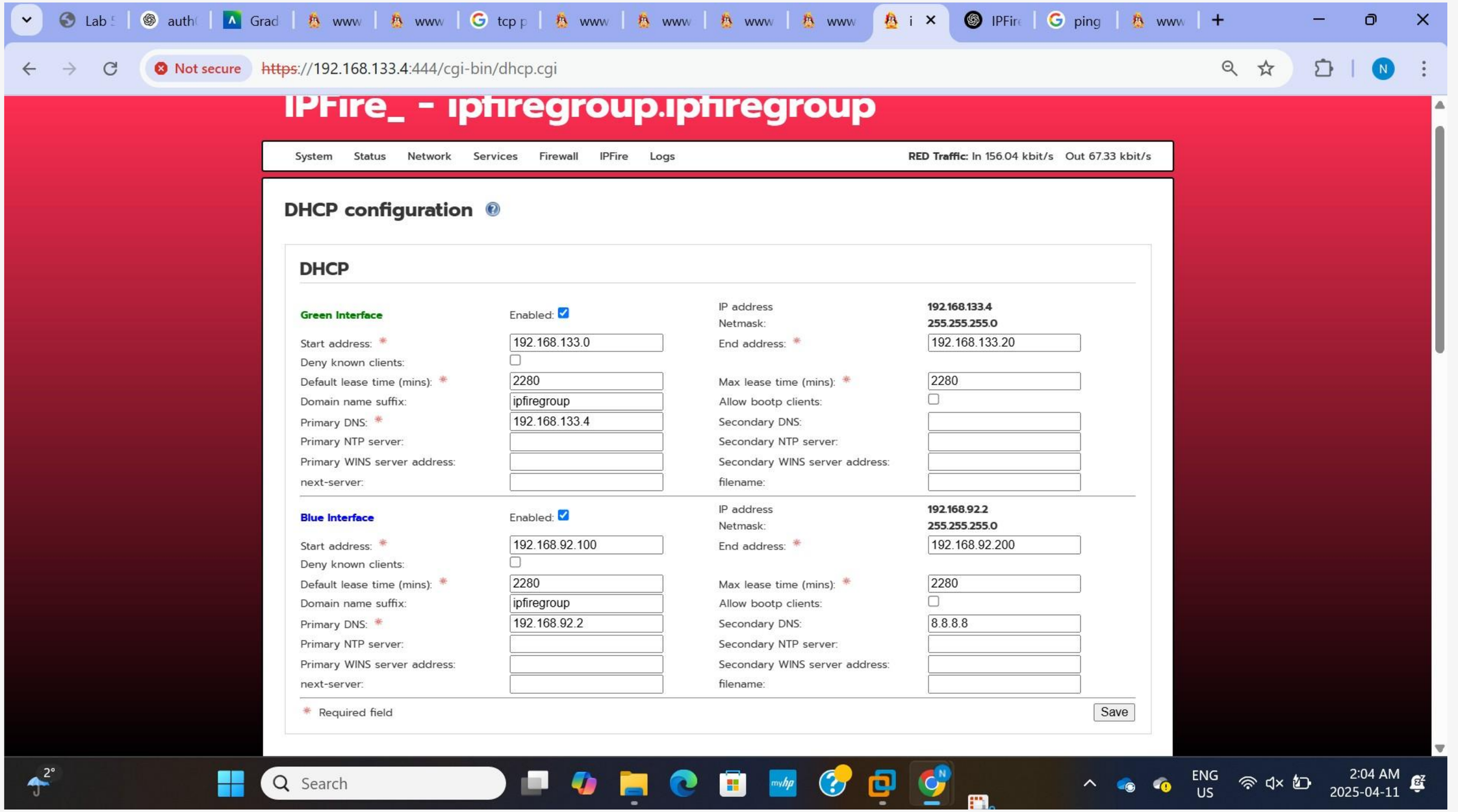
blue access


Not secure
https://192.168.133.4:444/cgi-bin/wireless.cgi
IPFire_ - ipfiregroup.ipfiregroup
System Status Network Services Firewall IPFire Logs
RED Traffic: In 243.57 bit/s Out 243.57 bit/s
Wireless Configuration
Add device
Source IP:
Source MAC Address:
Remark:
Enabled:
Required field
Add
Devices on BLUE
Hostname Source IP MAC Address Remark Action
192.168.92.0/24 NONE
nmagero-ubuntu.ipfiregroup 192.168.92.128 00:0c:29:47:5e:56 nmagero-ubuntu Added from DHCP lease list
Current DHCP leases on BLUE
IP address MAC Address Hostname Lease expires (local time d/m/y)
192.168.92.100 00:50:56:c0:00:01 NikitaHp 12/04/2025 10:24:08
192.168.92.128 00:0c:29:47:5e:56 10/04/2025 20:32:30
Search
ENG US
1:56 AM 2025-04-11

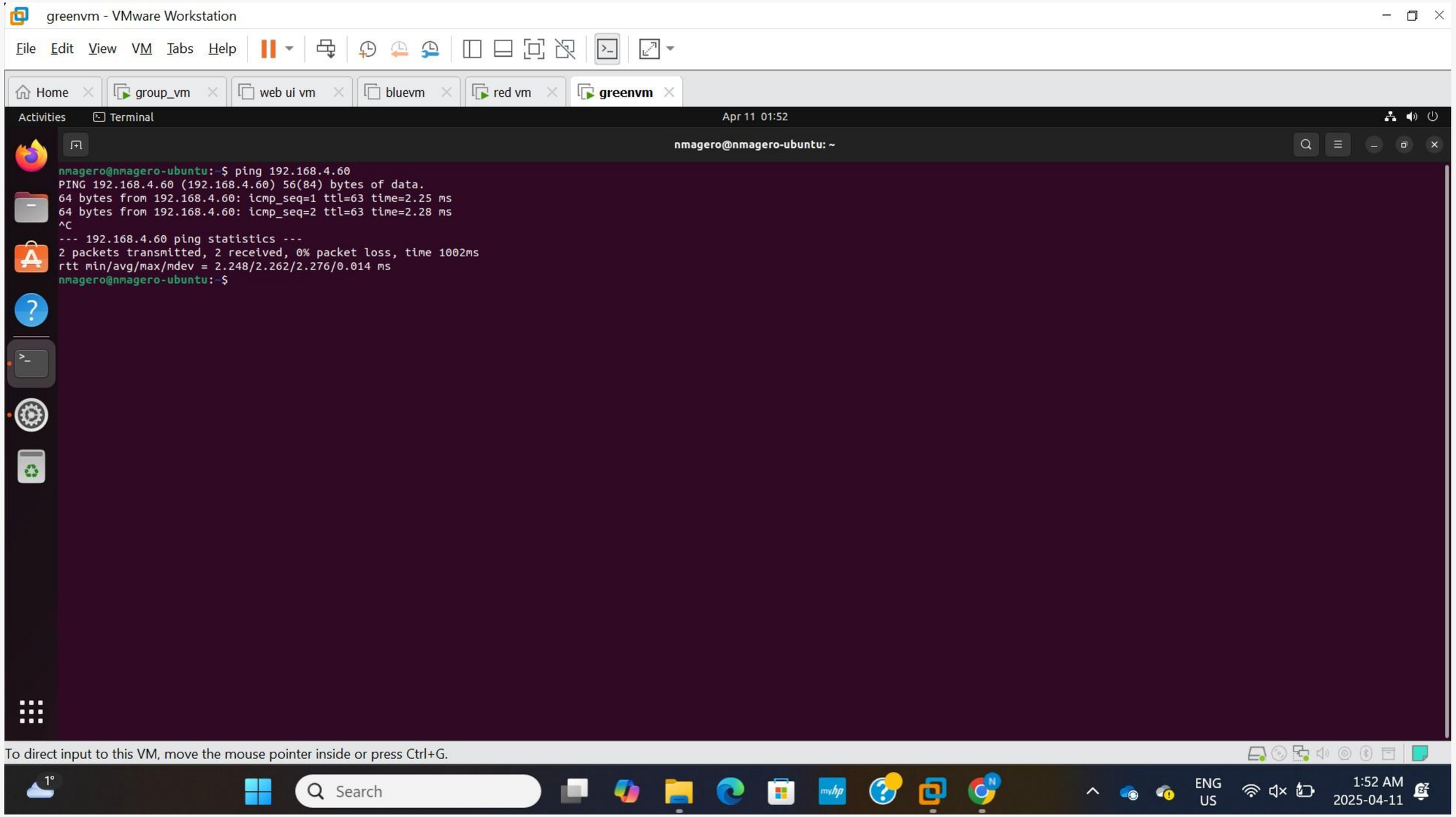
Firewall rules



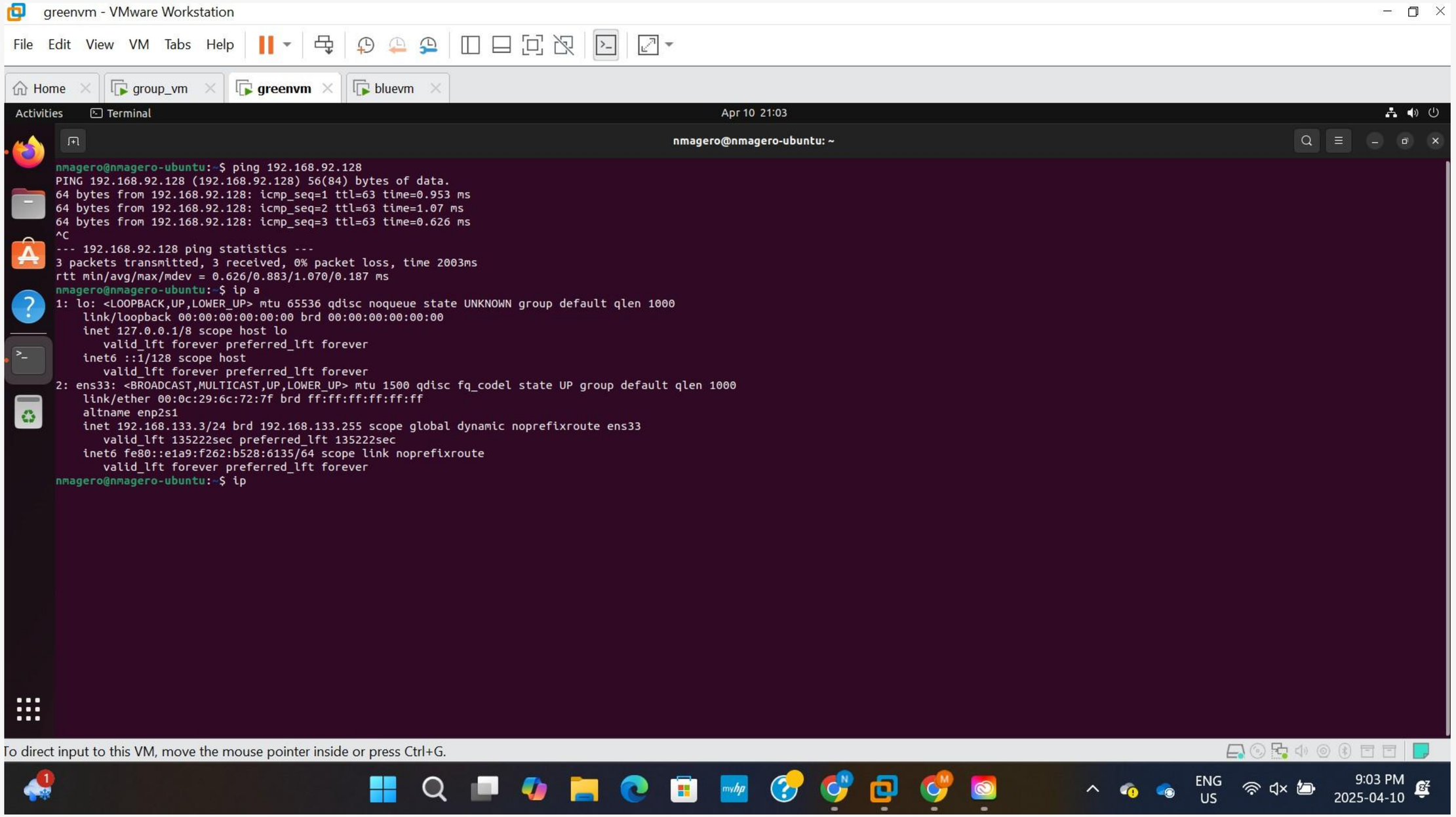
green configuration



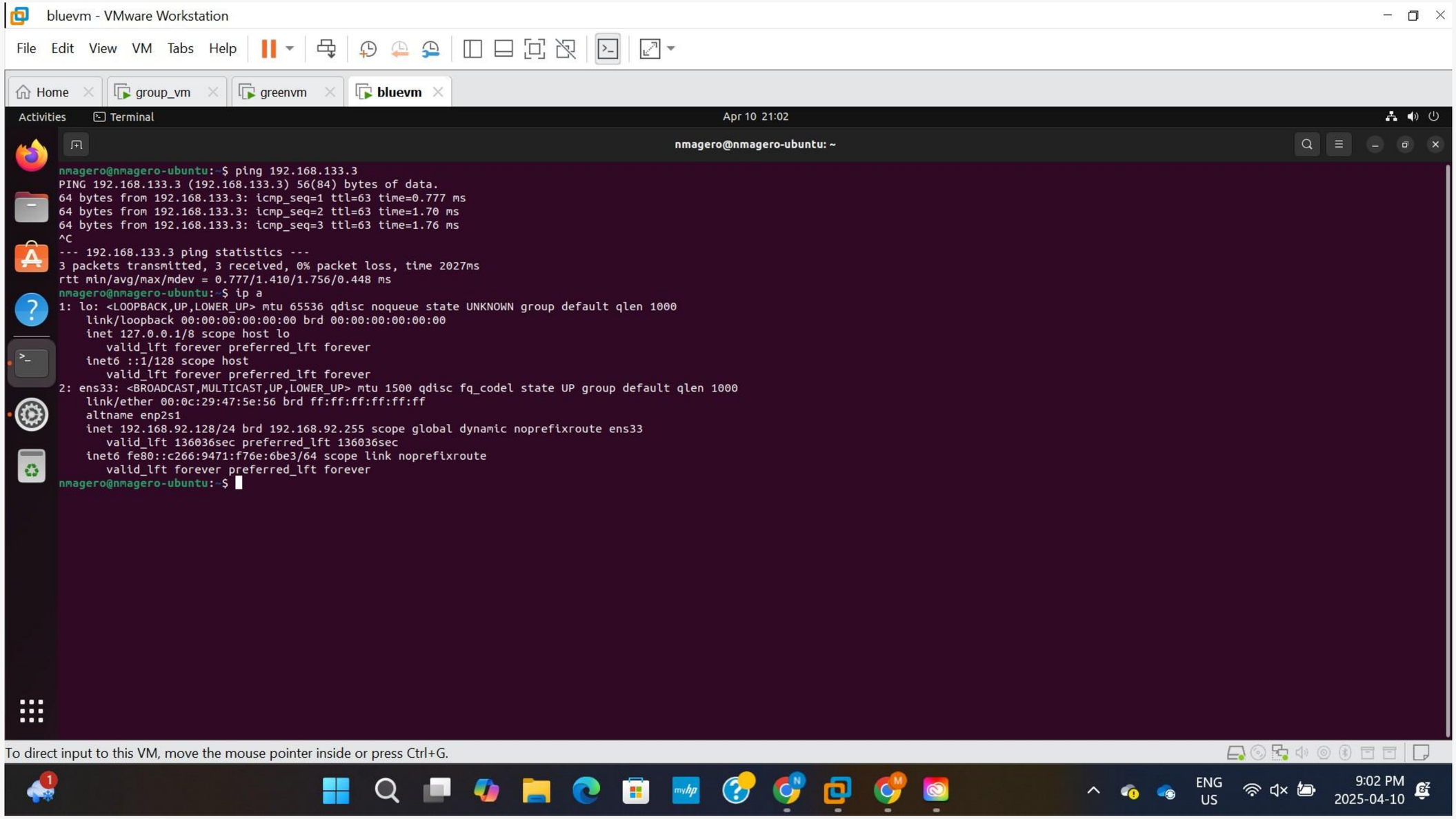
green pings red



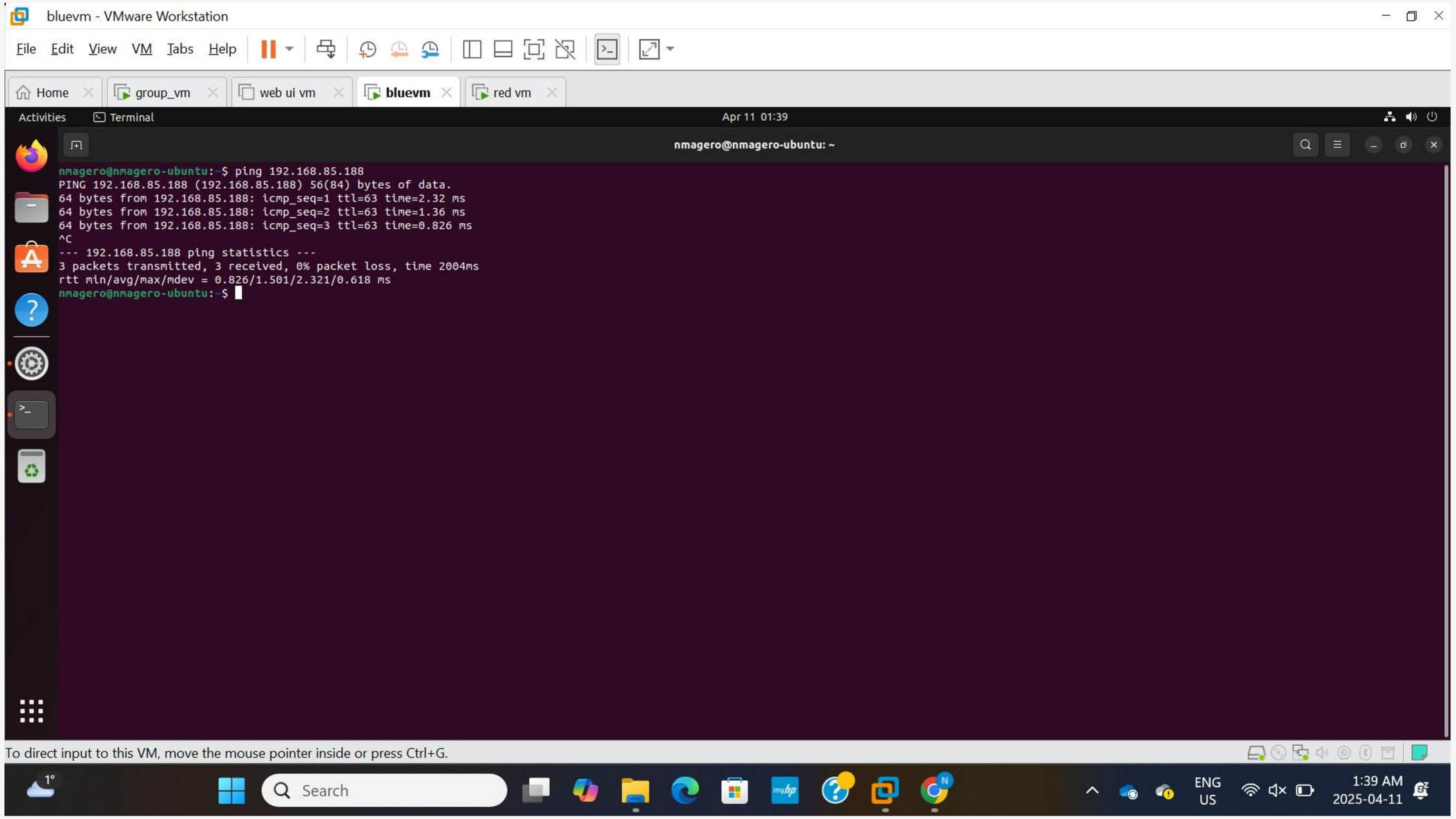
green pings blue



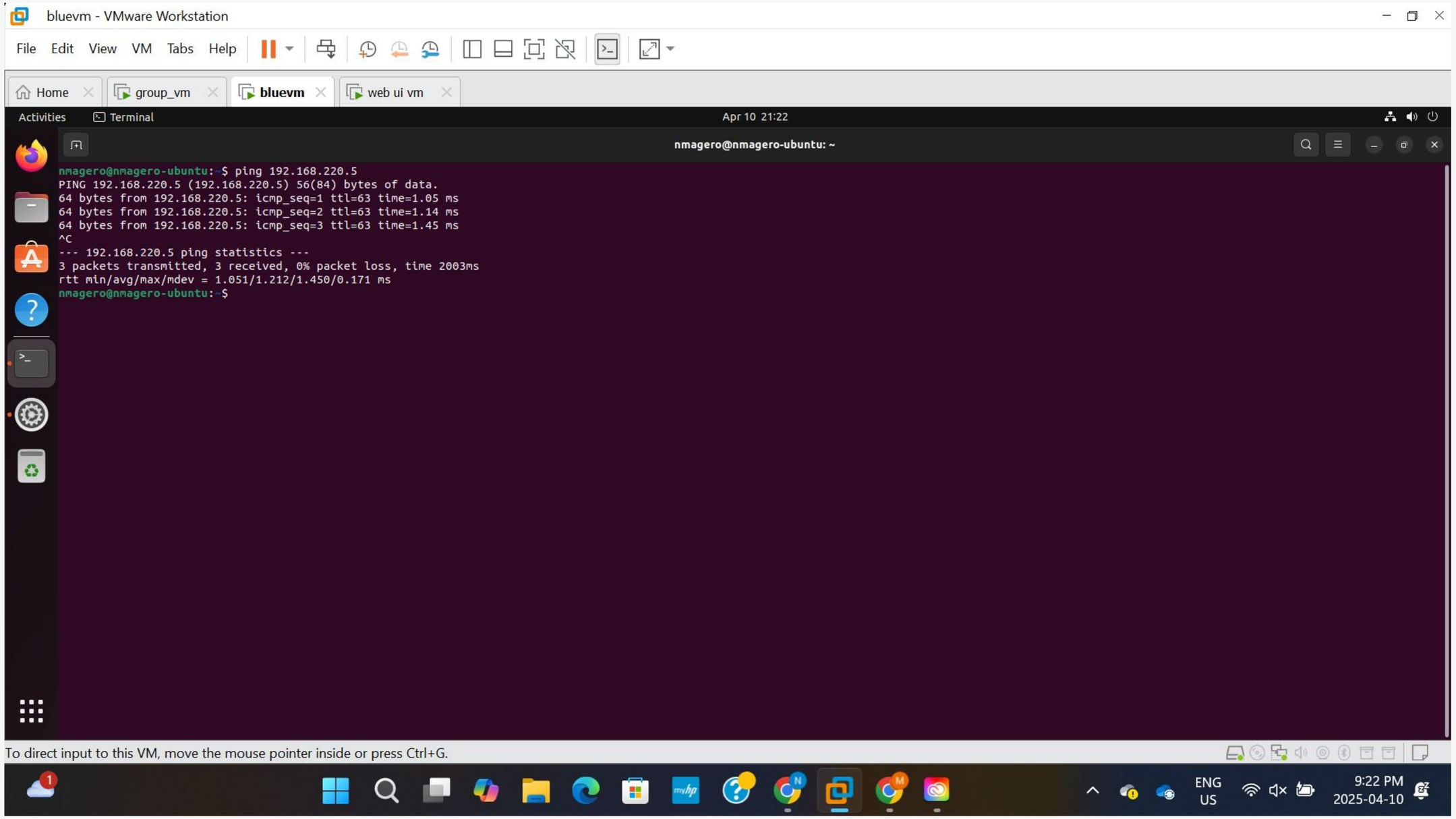
blue pings green



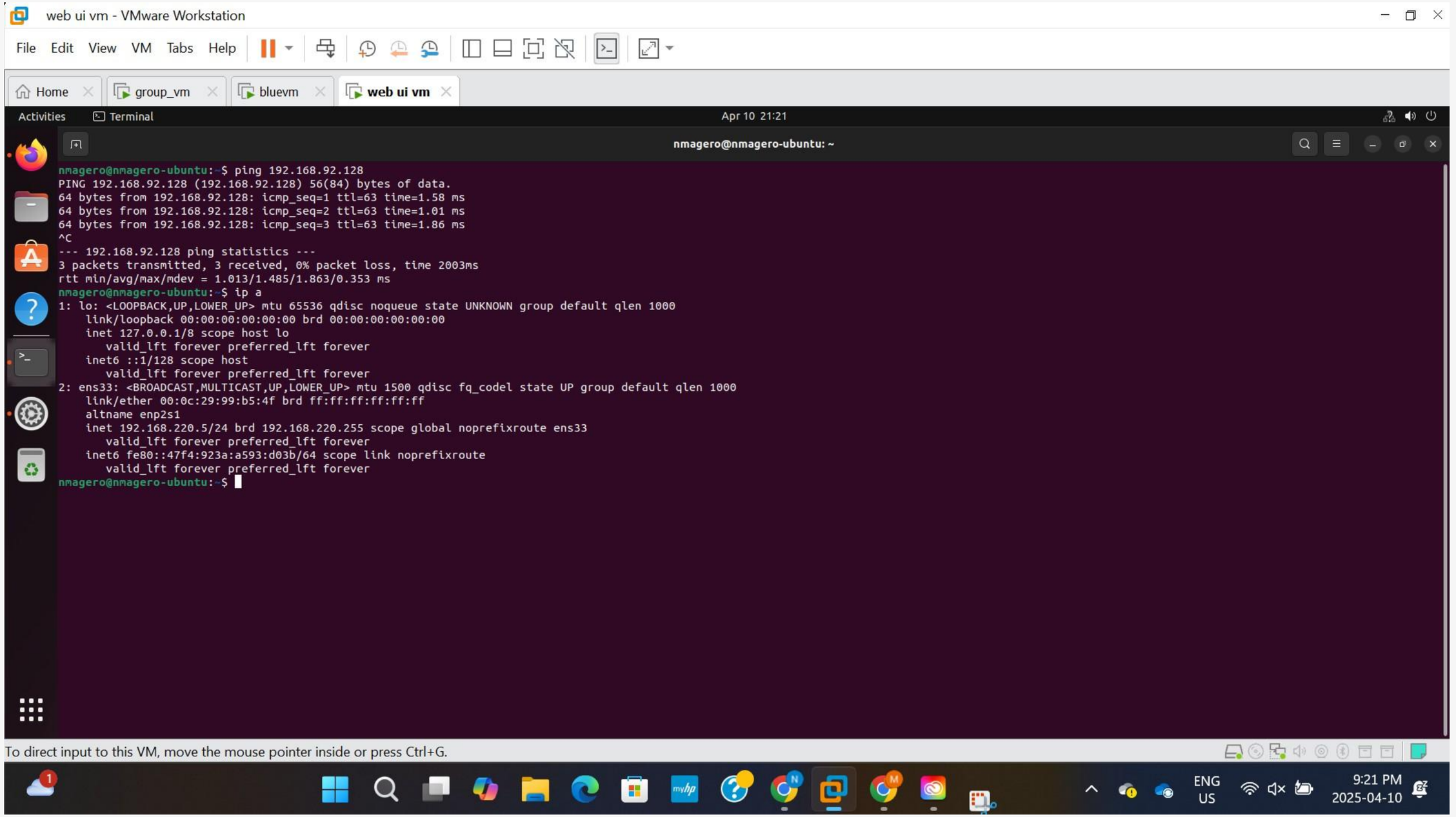
blue pings red



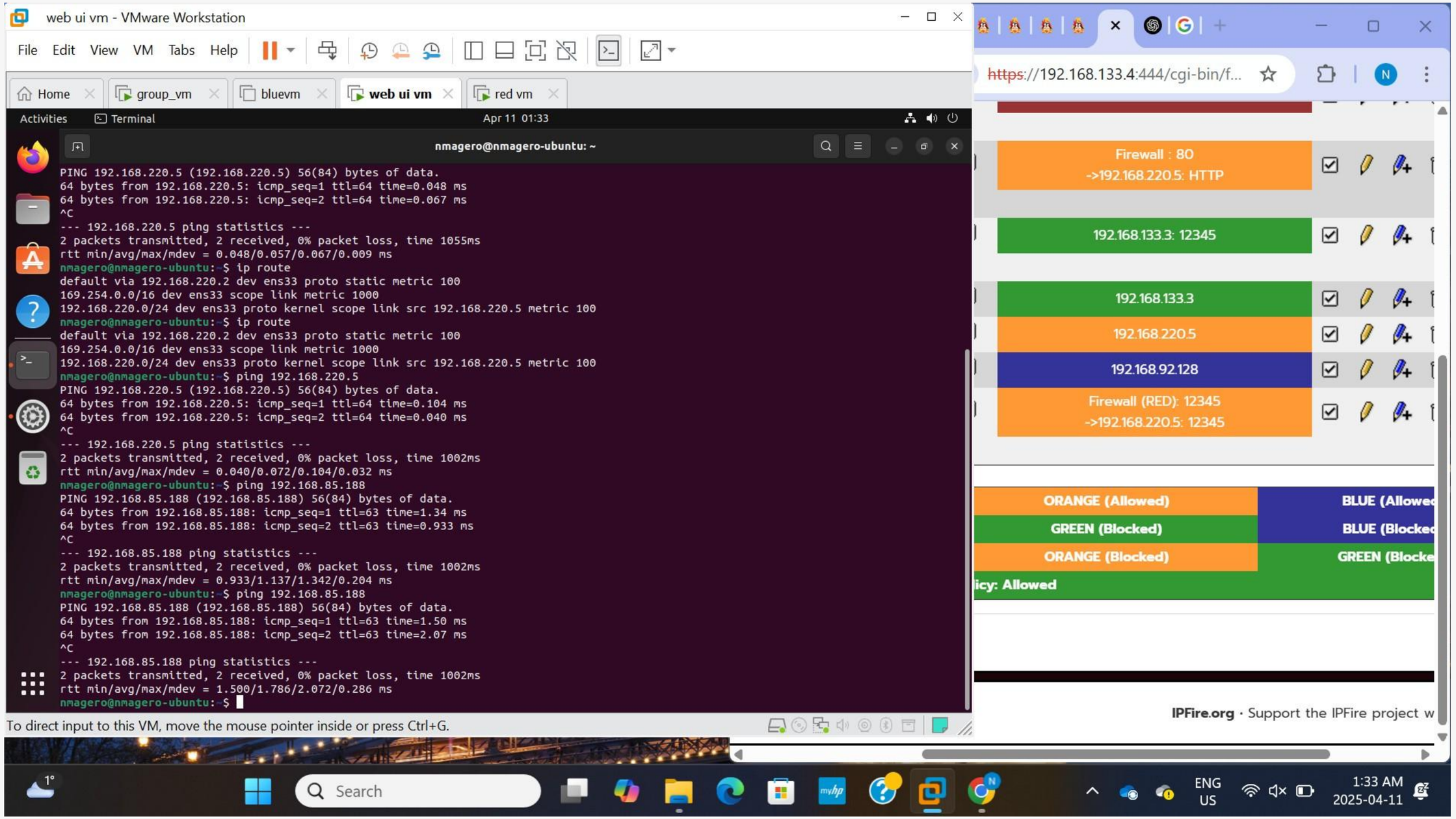
blue pings orange



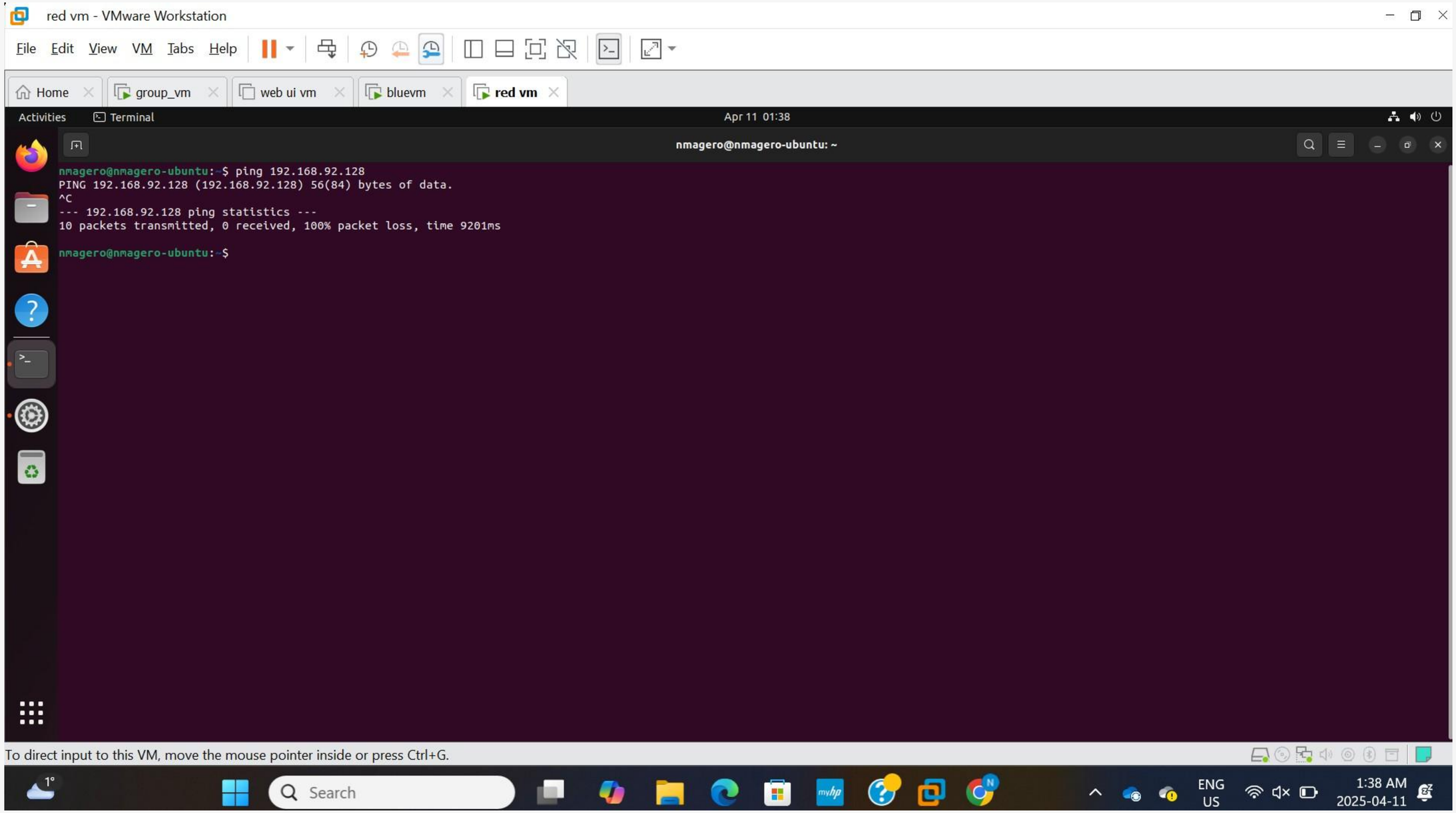
orange pings blue



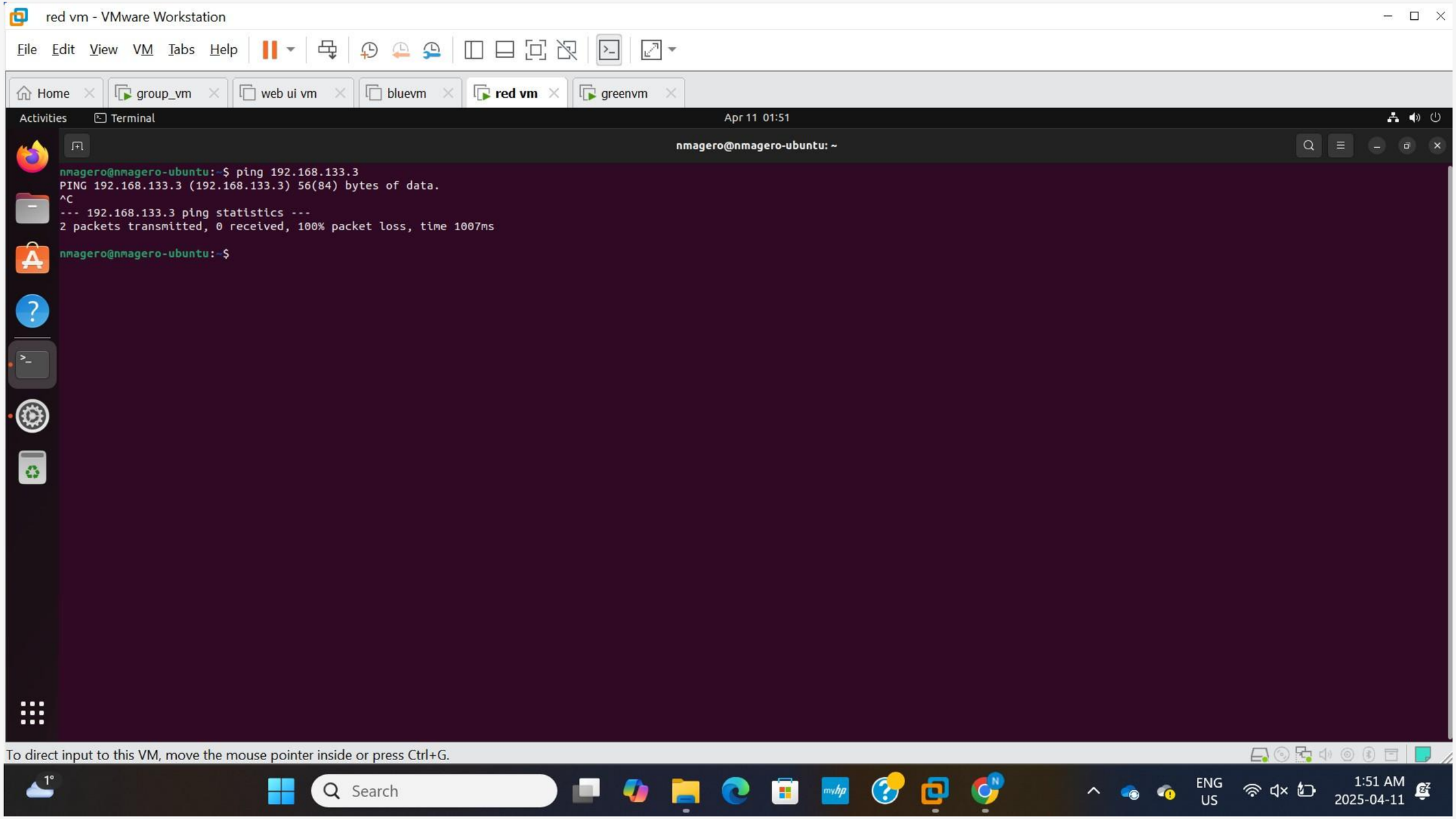
orange pings red



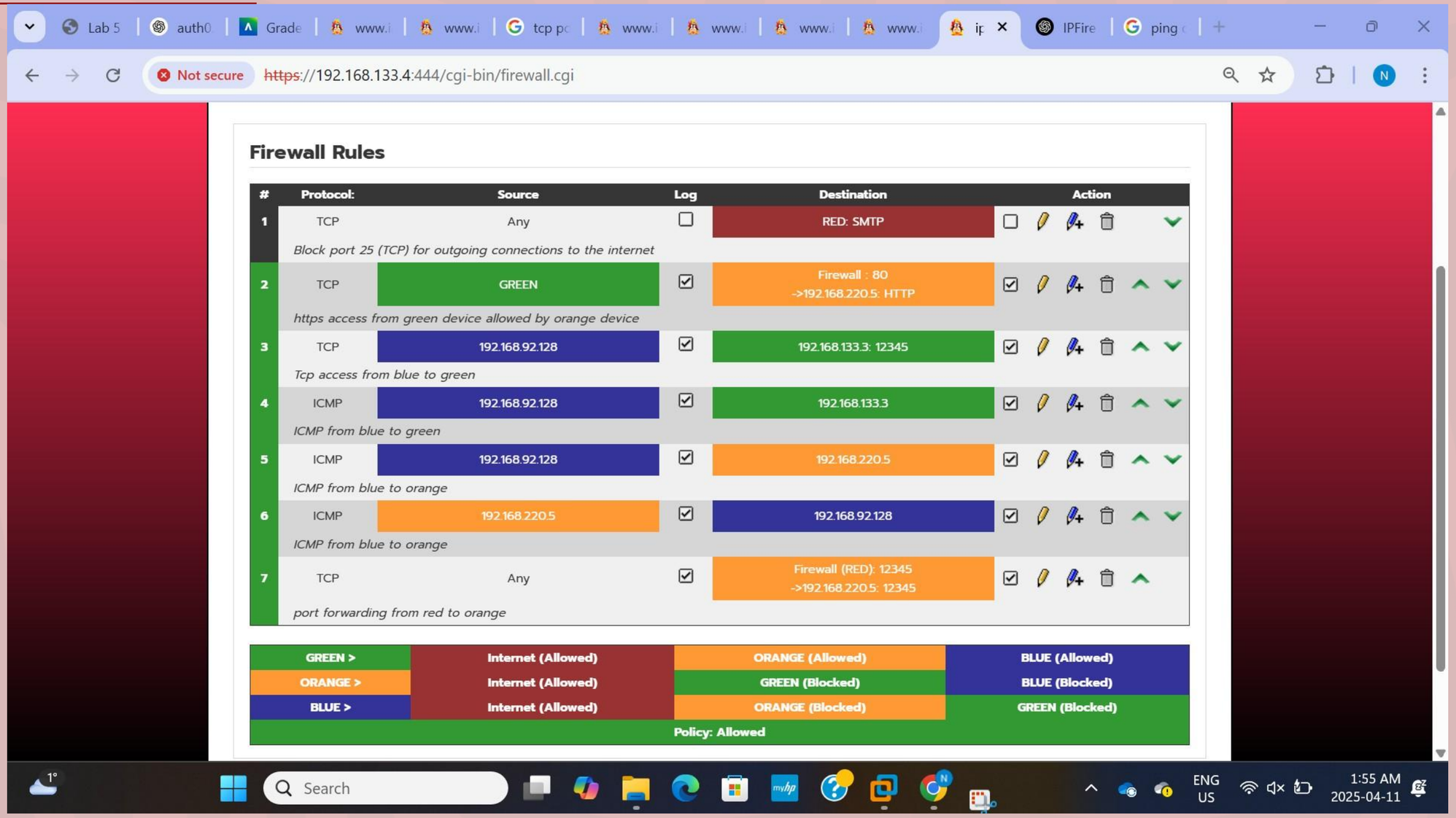
red cant ping blue



red cant ping green



Demonstration




thank
you

References/ Resources

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